DECO3500 · Design Proposal · Semester 2, 2026

NutriCart

Using Social and Context-Aware Technology to Support Healthier Grocery Choices

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The Problem Space

We are interested in understanding how social media influences the food choices of younger UQ students when grocery shopping in person around the St Lucia area, and how social and/or contextual computing could be used to support or influence these choices

Mostly Individual Focus

Most food-choice tools focus on the individual, with less focus on things like friends, peer influence and shared goals

Marshall et al., 2022

Inspiration vs Action

Social media can influence what people want to eat, but that doesn't always carry over into what they actually buy in the store.

Wang et al., 2024

Limited Social Support

Tools like ShoppingCoach can support individual choices in-store, but don't include much social interaction. Social nudges have mainly been explored online.

Strecker et al., 2024
Di Cosola & Neff, 2022

THE GAP: We couldn't find a system that incorporated social features, context-aware support and in-store grocery shopping together for university students at the same time.

Our Users & What We Found

Interviews with UQ students who grocery shop near St Lucia

Target Audience

→ UQ students aged 18-25

→ Shop at grocery stored in or near the Saint Lucia area (including Indooroopilly Shopping Center and Toowong Village).

→ Budget conscious because of student income

→ Students short on time due to study, work and social commitments

→ Living in share houses, student accommodation, or by themselves (i.e. not with parents).

Price is a big factor

"I ended up choosing the unhealthier option because of the price. Since I'm an international student, budget is tight."

Already using phones in-store

"If I'm in the grocery shop and didn't come prepared, I'll go on Instagram and search up easy, quick meals."

Personal recommendations

"My friend who goes to the gym cooked this really nice dish and told me the recipe. Two days later I immediately went to buy all the materials."

Timing matters

"I saw the reel right before I went out for groceries, and then I bought what was needed."

Design Opportunity & Concepts

How might context-aware mobile technology, combined with social features, support UQ students in making healthier food choices while grocery shopping near campus?

A Social Shopping Companion App

Uses location to recognise when a student is shopping. It could then show recommendations from friends, shared recipes and group goals while they are in the store.

B Recipe-to-Cart Bridge App

Saved recipes from social media could be turned into shopping lists. When the student is in-store, the app could remind them about the recipe and show ingredient options and prices.

C Housemate Meal Coordination App

Helps housemates plan meals together, organise who is buying what, and split larger purchases. This could also help reduce food waste and costs.

What exists & what's missing:

Personal fitness apps like MyFitnessPal focus mainly on the individual, the Woolworths app doesn't have a strong social component, and many meal-planning apps don't use location. We haven't found one that combines all three.

Plan of Work

What we still need to find out

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Wk 5-6 **Deeper Research**

Finish our interviews, shop with users to see how they actually grocery shop, and run a co-design session to narrow down our ideas.

Wk 7-8 **Low-Fi Prototyping**

Create prototypes to test with users and use their feedback to work out what the design needs.

Wk 9 **Stand-up**

Show our research findings, prototype progress and what we've learned from testing.

Wk 10–11 **Build & Test**

Build the interactive prototype with the main social and context-aware features.

Wk 12 **Tradeshow**

Live demo and pitch and test the prototype with users before the exhibit.

Wk 13 **Submit**

Refine the prototype and complete our final documentation and submission.

We want to understand what actually influences students' choices and what kind of support would be useful while they're shopping.